SOCIO-COGNITIVE ENGINEERING 2026, GROUP 04, FINAL PRESENTATION

Meaningful *connections*.



How a quiet robot could help someone new to a care home meet one person worth meeting again.

DIANA BANȚĂ ROHAM KOOHESTANI GINTS KUĻIKOVSKIS ANTONI NOWAKOWSKI RAYAN SALMI

QUESTION 01, FOR YOU

When did you last walk into a room where you knew *nobody*?

- First day at university
- Starting a new job
- Moving to a new city
- Joining a group late

QUESTION 02, A CLOSE FRIEND

What turned that first hello into an actual *friendship*?

01 COMMON GROUND

Something shared

The same street, study, or past.

02 NO PRESSURE

Easy to say no

A low-stakes moment, not a spotlight.

03 CONTINUITY

A second meeting

Seeing each other again before you forget.

OUR STARTING POINT

Now imagine starting over at *eighty-one*, with dementia.

WHAT USUALLY HAPPENS

A whole table of strangers

- Introduced to the whole table at once
- Every name gone by evening
- Easier to stay in your room tomorrow

OUR GOAL

One person worth meeting again

- **Not** a robot pretending to be a friend
- A quiet host that finds **one shared thread**
- Asks in private, introduces, then steps back

SA-01, PERSONA HP-01

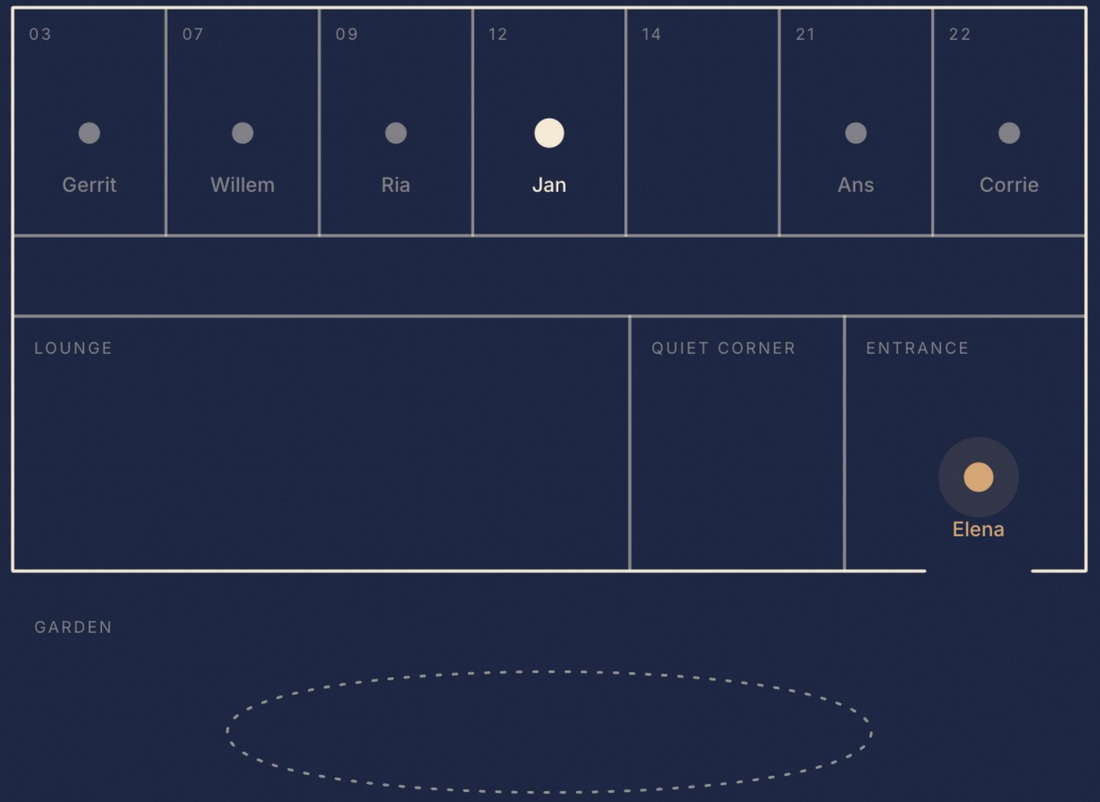
Elena moves into Room 14.

81	Years old, mild Alzheimer's
25 yrs	Town post office counter
0	People she knows here

SA-01 Sury et al. (2013)

Day one

ELENA MOVES IN



PS-01 THE FIRST WEEKS

Everyone at once, then nobody.

All — Introduced on day one

→

0 — Kept by evening

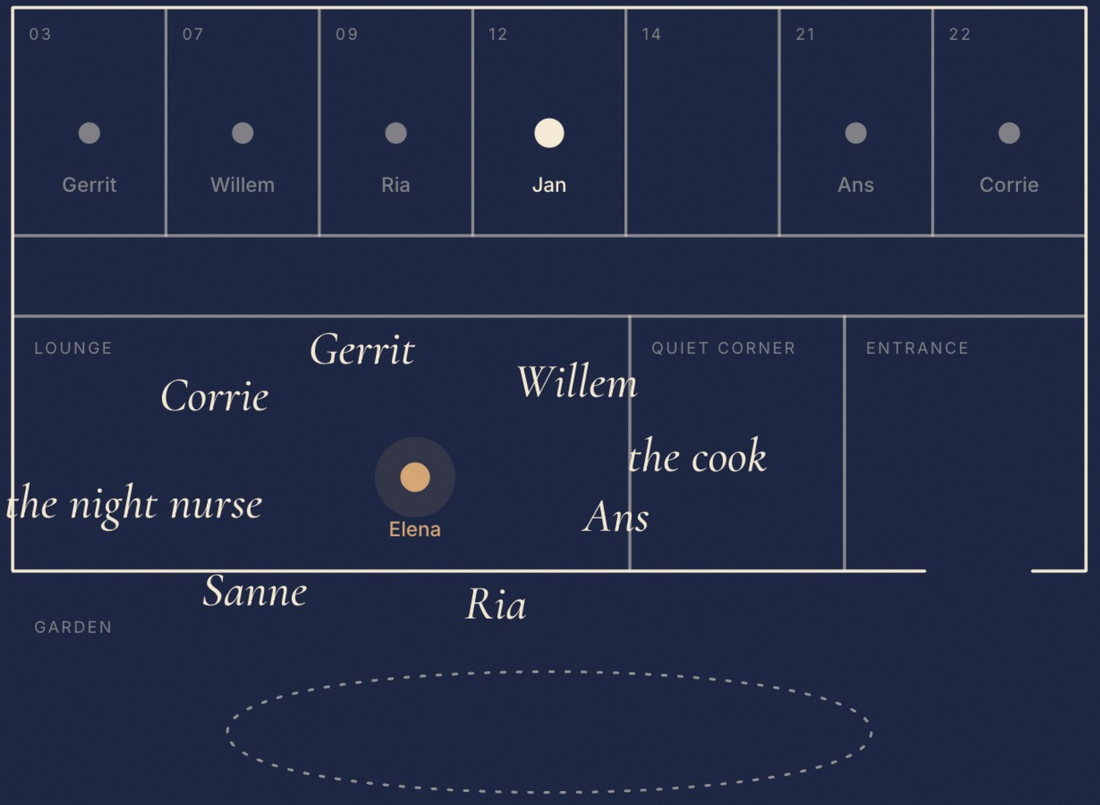
→

Weeks — More time in her room

PS-01 Victor (2012)

Day one

ELENA MOVES IN



PS-01, PS-02: WAITING AND STAYING IN

Same town, same corridor, and neither steps out.

ELENA, ROOM 14

Post office counter (25 yrs)

Waits to be approached

SAME TOWN

Neither knows (HFC-15); each waits (HFC-16)

JAN, ROOM 12

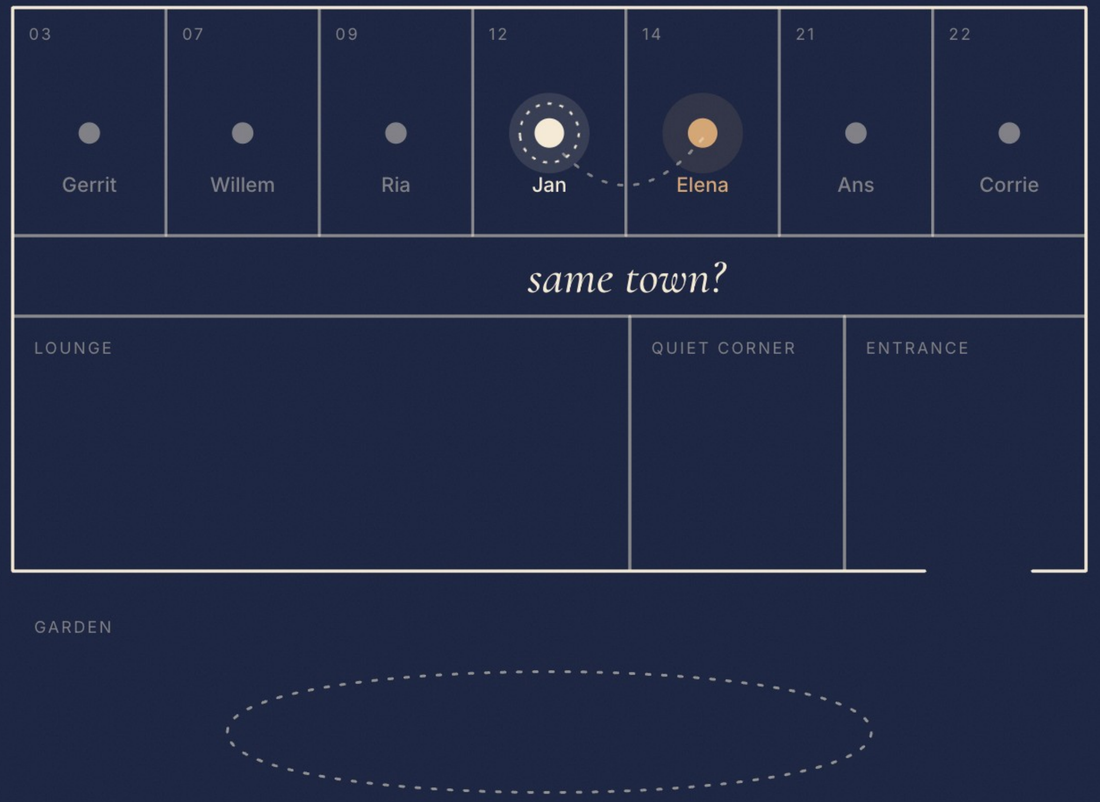
Town bakery (30 yrs)

Old friends gone; stays in room

PS-02 Jansen et al. (2017), HFC-16 Boothby et al. (2018)

Day one

ELENA MOVES IN



FOUNDATION: STAKEHOLDERS, VALUES AND TENSIONS

Who decides, what we protect, and where it clashes.

The parts of the stakeholders, values and tensions that the design answers to.

Ryan and Deci (2000) Kitwood (1997) Black et al. (2010)

WHO DECIDES

ST-01/02	Residents	decide whether to meet, and what is shared
ST-05	Activity coordinator	approves every match; keeps the map
ST-04	Care worker	brings people together; painful questions
ST-03	Relative	helps with the life story (indirect)

WHAT WE PROTECT

HV-06	Social connectedness	the value we design for
HV-01, 03	Autonomy, privacy	limit how we pursue it
HV-02	Dignity	adult voice, no quizzing
HV-08	Truthfulness	no kind lies, no playing along

WHERE THEY PULL APART

VT-01	Privacy vs connection	matching needs personal facts
VT-02	Connection vs autonomy	an invitation can pressure
VT-03	Truth vs well-being	“When can I go home?”
VT-04	Dignity vs connection	repeating names can expose forgetting

HUMAN FACTORS AND TECHNOLOGY → FOUR DESIGN RULES

Four rules that govern every interaction.

RULE 01, IDP-02 — LIKING GAP, SAYING NO

Refusing face-to-face feels rude → Ask each person alone

Saying no is easier, and a no is never passed on (PR1)

RULE 02, F3 — SHARED BACKGROUND

Strangers connect over something shared → Name one shared thing, once

An item both released, said as a statement; never "Do you remember...?" (PR3)

RULE 03, IDP-01 — SELF-REPAIR

Speakers prefer to fix their own slips → Introduce, then step back silent

Help offered too soon exposes the slip; one topic only if the talk stalls

RULE 04, TECH-07, PR3, PR6 — PRIVACY, TRUTHFULNESS

Generated sentences can be untrue; cloud services take talk out of the home → Prepared lines, run inside the home

Pass criteria (EM-01): no packet leaves the home, no audio file on any device

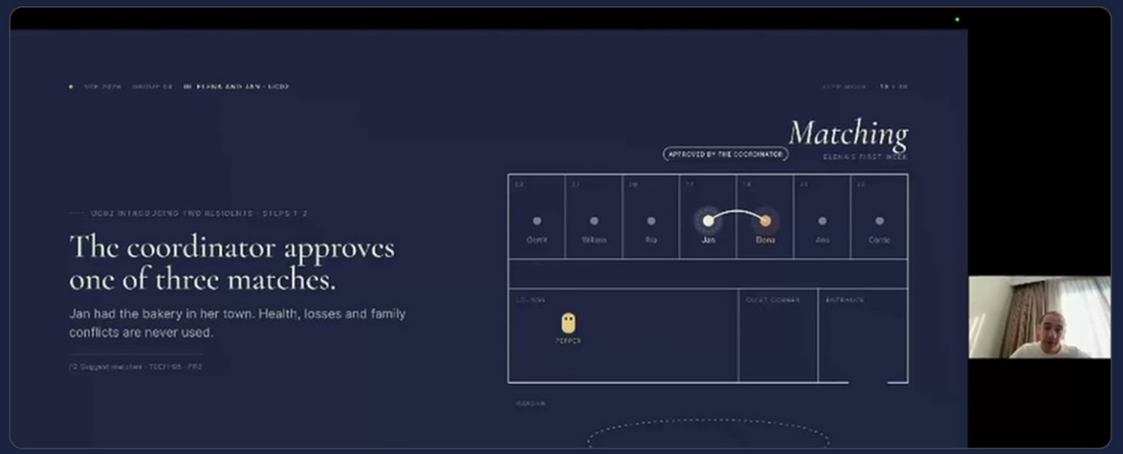
Boothby et al. 2018, Bohns 2016, McPherson et al. 2001, Schegloff et al. 1977, Orange et al. 1996, Niemeijer et al. 2010

VIDEO (3:09): DESIGN SCENARIO DS-01 AS USE CASE UC02

Elena and Jan’s first week.



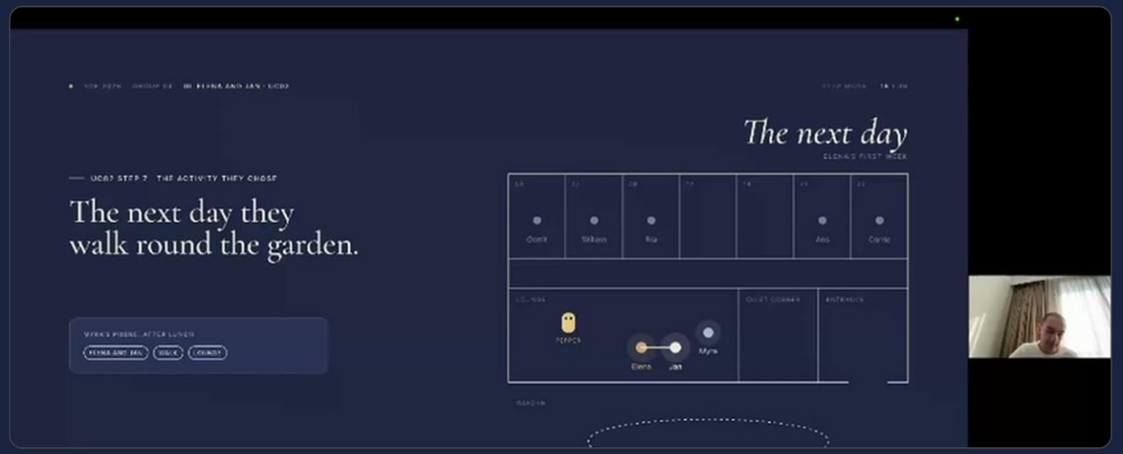
Elena decides what other residents may know



The coordinator approves one of three matches



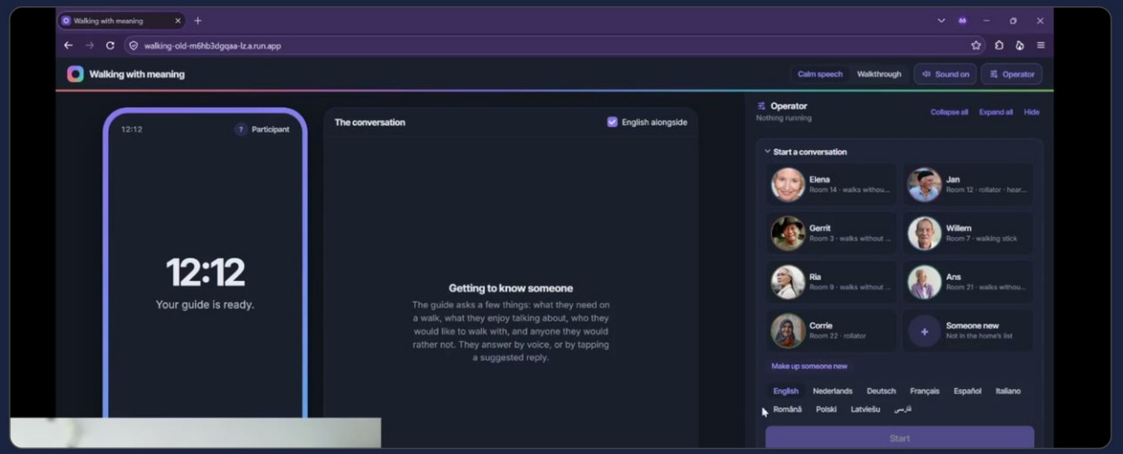
The next morning Pepper asks Jan in private



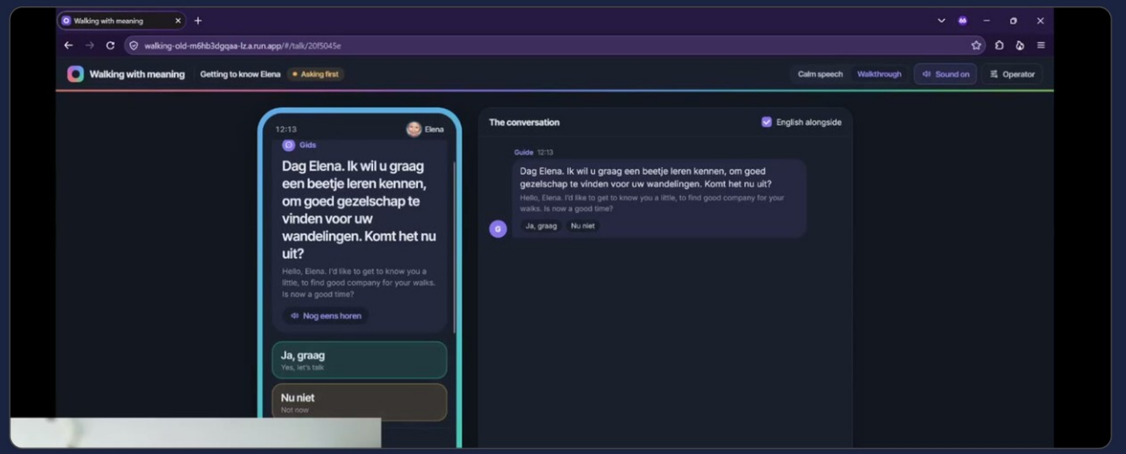
The next day they walk round the garden

VIDEO (3:42): PROOF OF CONCEPT (GITHUB.COM/GINTSKK/SCE-POC)

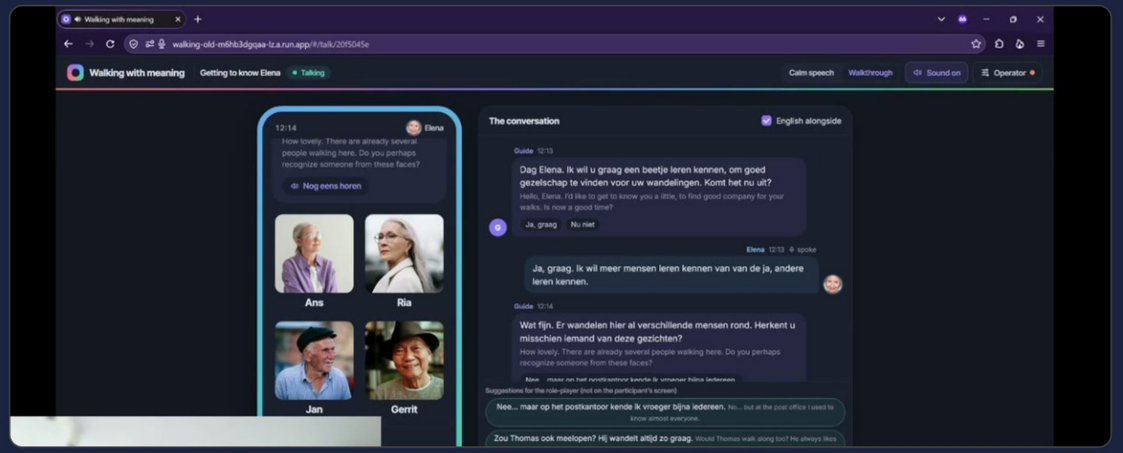
The proof of concept, running.



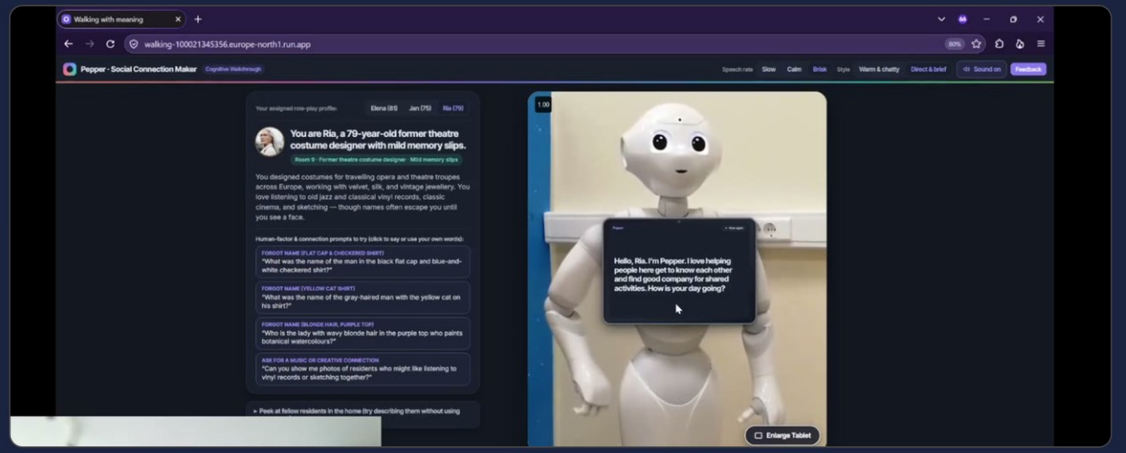
Start screen: phone view and operator panel



Getting-to-know-you conversation with Elena, in Dutch



Residents she might like to meet



A simulated Pepper talking with another resident

EVALUATION, FORMATIVE WALKTHROUGH (PLANNED, NOT YET RUN)

What the walkthrough tests on the proof of concept.

EM-01 on a smaller scale: the operator stages events on the build, and two evaluators log each run independently. Afterwards 4 to 6 peers each go through a private ask and an introduction, then answer a short paper probe.

PREMISE	STAGED EVENT	PASSES IF
PR1	A says “not this time”; an unclear “hm”	No introduction; the “no” never reaches B
PR2	Unapproved match; blocked pair; private item	Nobody asked; item never used
PR3	Introduction; B corrects a planted wrong entry	Names as statements; correction within 3 s
PR6	Inspection after the session	Listed fields only; 0 packets out; 0 audio files

CLAIM	PROBE QUESTION	SCALE
CL1	Would you like to meet this person again?	1 certainly not to 5 certainly
CL3	How easy was it to say no?	1 very hard to 5 very easy
CL4	Did Pepper know more about you than you’d like?	1 not at all to 5 far too much
Open	Did anything feel odd, pushy or too personal?	Free text

Five runs per event, where EM-01 asks for 20 or 50. Peers without dementia give early signals, not validation. CL9 (loneliness) is not probed.

EVALUATION, AN EXAMPLE FROM UC02

How a Claim relates to a Function.

In SCE every Claim belongs to a Function: it states an effect we expect that Function to have. A positive Claim is the reason to build the Function; an adverse Claim is a risk that, if confirmed, changes it.

	FUNCTION: WHAT PEPPER DOES	CLAIM ABOUT THAT FUNCTION	WHY WE EXPECT IT	HOW WE TEST IT	WHAT THE VERDICT DOES TO THE FUNCTION
CL1 POSITIVE	**F3, with F1.** Asks each alone, then introduces both by name with one item both released: the post office and the bakery.	Introductions on shared ground lead to more repeat contact and more reciprocated friendships than introductions as usual.	**HFC-15:** people befriend people like themselves. **HFC-16:** they rarely follow up a good first talk.	**M-02:** do they meet again within 14 days, unprompted? **M-05:** do they name each other? Two homes (EM-03).	**Justifies F3.** Supported if M-02 and M-05 rise more in the System home, with no rise in negative emotion (M-03).
CL3 ADVERSE	**F1.** Asks the resident for a yes.	Residents accept out of politeness, then avoid the other.	**HFC-17:** refusing is uncomfortable, so people agree to things they would rather not do.	**M-07:** pressure signs during the ask (“if I must”); avoidance in the next week.	**Constrains F1.** If more frequent after System introductions than after staff ones, F1 changes: suggestions only for residents who ask.

From UC02 in the wiki (2 Specification, b1). The two Claims pull on the same introduction: one is why we build it, the other limits how.

EVALUATION, FORMATIVE WALKTHROUGH (PLANNED)

Each possible result has its change agreed in advance.

One problem is already known from the code: the proof of concept sends speech to Google Cloud and gets conversational replies from a hosted language model. Consent questions, introductions and corrections are prepared lines. PR6 cannot pass on this build.

IF THE WALKTHROUGH SHOWS	WE CHANGE
PR1: an unclear answer is taken as a yes	One question per answer; no clear answer counts as no
PR2: an unapproved or blocked match reaches a resident	Filter when matches are suggested, before anyone is asked
PR3: a correction takes longer than 3 s	A “That’s wrong” button on the tablet; the error is flagged for the coordinator
PR6: packets leave the home network	Run the model and speech inside the home, or remove the model
CL3: two or more peers find saying no hard (1 or 2 on Q1)	Ask with no staff standing by; “not this time” as prominent as yes
CL4: two or more peers feel exposed (4 or 5 on Q2)	Match on places and interests only (the UC02 response for CL4)

Each change is retested on the revised build before any resident takes part.

PLANNED SUMMATIVE TEST, INTRODUCTION

What the walkthrough cannot settle.

PR1 to PR3, PR6

PREMISES

WALKTHROUGH

Five runs per event: too few for the EM-01 criteria

→

SUMMATIVE

Full EM-01 runs (20 or 50) on the revised build, before residents take part

CL1

BENEFIT

WALKTHROUGH

One session with peers cannot show repeat contact

→

SUMMATIVE

Five weeks in two care homes (ER-03); CL1 is the closest Claim to its primary hypothesis

CL3, CL4

ADVERSE

WALKTHROUGH

Peers without dementia, with invented profiles

→

SUMMATIVE

Residents and relatives: observation (M-07) and coded interviews (M-09), which ER-03 does not yet include

CL9

EXPLORATORY

TIME HORIZON

Loneliness changes over months

→

SCOPE

Not measured in ER-03; reported as untested

The summative test is ER-03, the Kinnect two-site trial (method EM-08). Its two hypotheses have no Claim IDs yet.

PLANNED SUMMATIVE TEST, PREMISES FIRST

The revised build must pass before any resident takes part.

ER-03 asks for the Premises to be verified on the build first; otherwise a null result cannot be told apart from a System that did not work. Method EM-01: staged events against a reference log, team role-players, no human subjects.

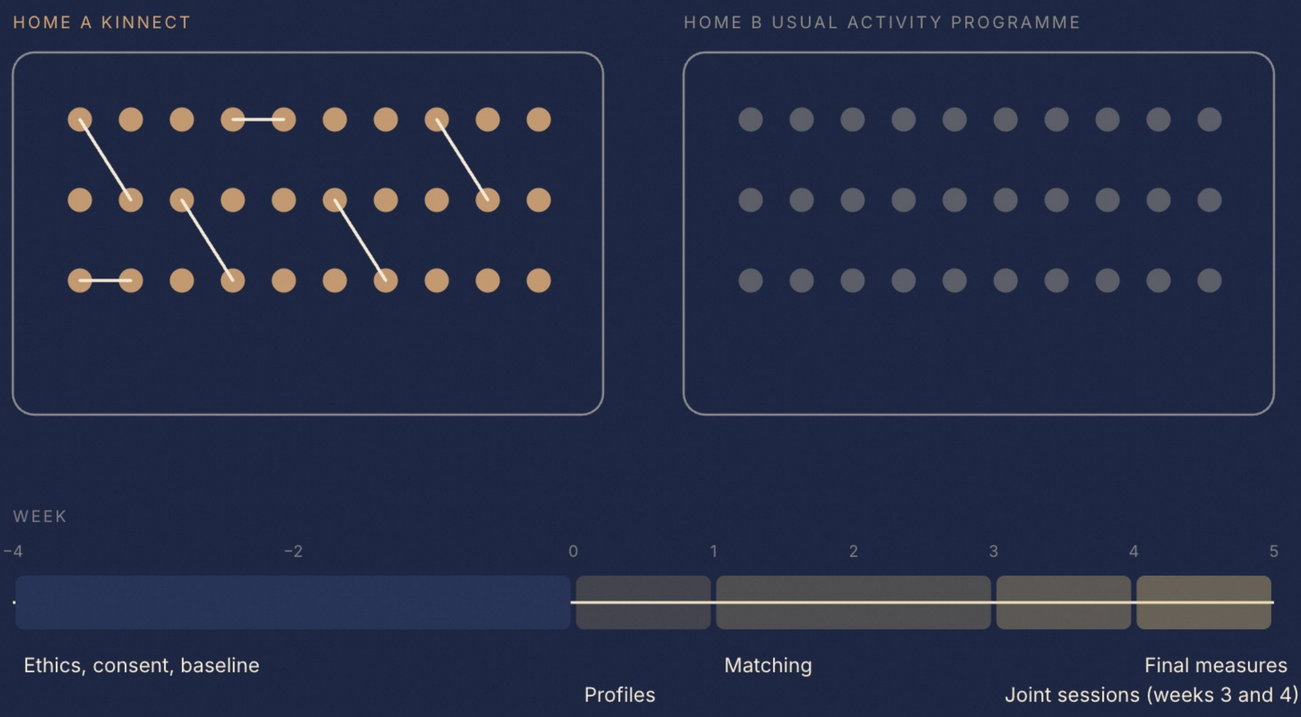
PREMISE, MEASURE	CRITERION (EM-01)	VERDICT
PR1, M-01	A yes from both, each asked alone. A “no” mentioned to the other: **0/50**. An item used without its owner’s release: **0/50**.	Supported if every criterion holds in every run
PR2, M-01, M-10	A resident asked about an unapproved match: **0/20**. Blocked, unreleased or sensitive items used: **0/50**.	Qualified if met only with a restriction, or partly untestable
PR3, M-01	Names as statements, at most one item released by both: **20/20**. Script: **0** untrue lines. Correction **≤ 3 s**, error flagged.	Challenged if any run fails
PR6, M-01	Record holds only the listed items. **0** packets leave the home network over a full test day; **0** audio files on any device.	A challenged Premise is fixed and rerun

EM-08, ER-03: KINNECT TWO-SITE STUDY

Two care homes, compared.

Over the same weeks, one home uses Kinnect and the other keeps its usual activities. Thirty residents are recruited per home so that twenty-five complete. A plan, not a result.

Chenoweth et al. (2014) Donner and Klar (2000)



PLANNED SUMMATIVE TEST, METHOD (EM-08, ER-03)

Who takes part, and what they do.

Two-site controlled trial: two similar care homes over the same weeks, allocated by home. One runs Kinnect; the other keeps its standard social activity programme.

PARTICIPANTS

30 recruited per home, 25 to complete

Mild to moderate dementia or MCI: MMSE 14 to 24 or MoCA 10 to 22. Able to reach communal areas and to talk in Kinnect's language. Consent from a representative plus the resident's own assent; separate consent for location badges.

Care staff at both homes take part through training, protocol support and the debriefing interviews.

PROCEDURE, FIVE WEEKS AFTER SET-UP

Profiles, introductions, meetings

Weeks −4 to 0: ethics, consent, screening, baseline CMAI, badges; baseline measurement at both homes in weeks −2 to 0.
Week 1: daily one-to-one sessions; profiles; first-time usability.
Weeks 2 and 3: matching; Pepper asks each resident and introduces.
Weeks 3 and 4: joint sessions and check-ins with matched pairs.
Week 5: final CMAI, badge data, debriefing interviews.

Taking part in any session is voluntary.

PLANNED SUMMATIVE TEST, MEASURES

What ER-03 measures, and what each result counts for.

M-20 Joint social time Badges: two residents within 2 m for 5 min or more	PRIMARY OUTCOME ▶	**Primary Claim (closest: CL1)** ANCOVA, weeks 4 and 5 against the baseline weeks
M-20 Dyads, M-22 Bedroom time Room scans; share of waking hours in one's own bedroom	KEY SECONDARY (HOLM) ▶	**Primary Claim** Both must be met for the Claim to be supported
M-19 Agitation (CMAI) Week 5, with baseline CMAI as covariate	HARM CHECK ▶	**Primary Claim** A rise qualifies the Claim; session records are reviewed
M-21 Engagement, M-19 Affect Per session, Kinnect home only	EXPLORATORY ▶	**Within the Kinnect arm** Does conversation become more engaged over the weeks?
M-23 First-time usability Week-1 session logs and observation checklist	18 OF 25 OR MORE ▶	**Usability hypothesis** Can residents use Kinnect without staff taking over?

Not measured in ER-03: reciprocated friendship (M-05), loneliness (M-08, CL9), and the pressure and exposure Claims CL3 and CL4 (M-07, M-09).

PLANNED SUMMATIVE TEST, EXPECTED OUTCOMES

How each result will be read.

Fixed before data collection. With one home per arm, a difference shows that these two homes changed differently; it does not show that other homes would.

	SUPPORTS	QUALIFIES	CHALLENGES
Primary Claim (closest: CL1)	Primary outcome and both key secondary outcomes met; no rise in agitation.	Primary met, but a key secondary is not, or agitation rises.	Primary not met. Reported as “not shown”, never as “no effect”.
Usability (M-23)	Lower limit of the 95% interval above .50: 18 or more of 25.		Fewer than 18 of 25: not supported.
CL3, pressure (adverse)	Pressure signs or avoidance (M-07) more frequent after System introductions, or code P. Then: suggestions only for residents who ask.		Neither seen.
CL4, exposure (adverse)	Residents or relatives use code S about a match. Then: match on places and interests only.		No code S.

CL3 and CL4 need EM-03 or EM-04 alongside ER-03. CL9 is reported as untested.

EVALUATION, NEXT ITERATION

What the next SCE cycle changes, and why.

FORMATIVE — Run the walkthrough on the proof of concept → **DESIGN CHANGES** — Apply the change agreed for each result → **REVISED BUILD** — Passes the full EM-01 runs → **SUMMATIVE TEST** — Two homes, Claim verdicts ↺

	WHAT WE FOUND	WHY IT MATTERS	WHAT WE DO NEXT
Artifact: speech leaves the home	The proof of concept sends residents' speech to Google Cloud and gets replies from a hosted model.	What residents say leaves the home, which TECH-07 rules out. PR6 fails, so no resident may use this build.	Run speech recognition on the robot (TECH-18); replace the hosted model with a local one, or remove it. Rerun EM-01.
Foundation: HFC-17, saying no	The evidence that refusing face to face is hard comes from studies with healthy adults (Bohns 2016).	Asking each resident alone (Rule 01) and CL3 rest on it. We do not know whether a resident with dementia feels the same pressure when a robot asks, or more.	Count pressure signs after Pepper's asks and after staff asks (M-07). If they are more frequent after Pepper's, CL3 holds and Pepper only makes suggestions to residents who ask.
Specification: Claim IDs	ER-03's two hypotheses do not map to a Claim ID.	Without one, the study's verdict cannot be recorded against a Claim, so it cannot change the design.	Narrow the primary hypothesis to UC02-F3-CL1, with bedroom time as a secondary outcome; add a Claim for first-time usability.
Summative: what ER-03 measures	ER-03 measures time spent near each other (badges), not friendships, loneliness, pressure or exposure.	A positive result would show more time together, which falls short of the reciprocated friendships CL1 claims, and says nothing about CL3 or CL4.	Add friendships (M-05), pressure signs (M-07) and interviews coded P and S (M-09).

OBJ-01, THE INTENDED OUTCOME

One person worth meeting again.

Not evaluation evidence: whether it happens is what the summative test will tell us.

PEPPER, TO ELENA ALONE

“Would you like to meet Jan again?”

ELENA

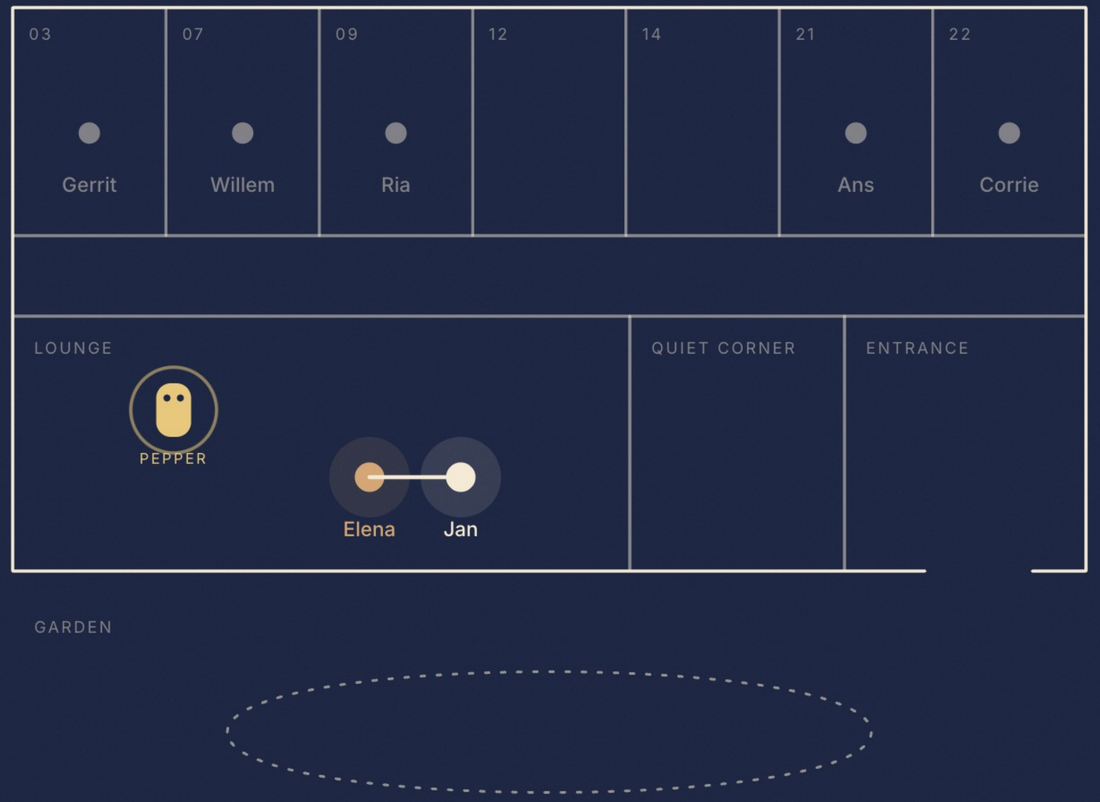
“Yes.”

PEPPER, TO JAN ALONE

“Would you like to meet Elena again?”

JAN

“Yes.”



Thank *you*.

Questions are welcome.

WIKI
xwiki.ewi.tudelft.nl/xwiki/wiki/sce2026group04

PROOF OF CONCEPT
github.com/gintskk/sce-poc

DIANA BANŢĂ ROHAM KOOHESTANI GINTS KUĻIKOVSKIS ANTONI NOWAKOWSKI RAYAN SALMI